



University of Toronto-Mississauga
CSC363 Final Examination
Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

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Instructions

1. Number of exam questions: 11
2. The full points here are 70. *You need to score at least 28/70 not to fail this course.*
3. Write your answers in the spaces provided. If you must use more room, use the space available on the last three pages, and indicate this on the question page where you ran out of room.
4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.



Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

(c) (3 points)

Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.
- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.
- (c) A set is recognizable if and only if it is c.e. (computably enumerable).
- (d) $P \subseteq NP$.
- (e) Every Boolean formula is equivalent to a 3CNF formula.
- (f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.
- (g) The Halting Problem is an NP-complete problem.

**Q5. Answer each of the following**

(a) (1 point)

Give an example of a set A of natural numbers which is both c.e. and co-c.e.
(co-c.e. means the complement of a c.e. set).

(b) (1 point)

Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)

Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

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#1 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?



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#1 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -*clique* if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#2 Page 6 of 12

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Q7. Let A, B , and C be three sets of integers.

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Q8. (5 points)

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(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#3 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

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#3 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#4 Page 10 of 12

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University of Toronto-Mississauga
CSC363 Final Examination
Computational Complexity and Computability Winter 2023

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6. Materials allowed: No additional materials are allowed.



Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \sqcup q_1 \ 1 \ R$$

$$q_1 \sqcup q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

(c) (3 points)

Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \sqcup q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

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**Q5. Answer each of the following**

(a) (1 point)

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(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

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#5 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#5 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#5 Page 10 of 12

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#6 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

- (c) A set is recognizable if and only if it is c.e. (computably enumerable).

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For A, B above: Is $A \leq_T B$? Justify your answer.

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#6 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

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#6 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

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(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Prove that $3SAT \leq_p L$.



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#6 Page 10 of 12

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Prove the following statement: Every finite language is decidable.

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#7 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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(b) (3 points)

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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#7 Page 8 of 12

Q8. (5 points)

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Prove that $3SAT \leq_p L$.



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#7 Page 10 of 12

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#8 Page 6 of 12

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#8 Page 8 of 12

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Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -*clique* if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#8 Page 10 of 12

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#8 Page 12 of 12

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University of Toronto-Mississauga
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$$q_0 \ 1 \ q_0 \ 1 \ R$$
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$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

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#9 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

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#9 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

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$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

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#9 Page 10 of 12

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Prove the following statement: Every finite language is decidable.

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#10 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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#10 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#10 Page 10 of 12

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#11 Page 10 of 12

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#12 Page 6 of 12

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#12 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -*clique* if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#12 Page 10 of 12

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University of Toronto-Mississauga
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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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$$gn(q_0 \sqcup q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



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#13 Page 6 of 12

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Q7. Let A, B , and C be three sets of integers.

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#13 Page 8 of 12

Q8. (5 points)

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#13 Page 10 of 12

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Prove the following statement: Every finite language is decidable.

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Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

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For A, B above: Is $A \leq_T B$? Justify your answer.

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#14 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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#14 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#14 Page 10 of 12

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#15 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

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#15 Page 10 of 12

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#16 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#16 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -*clique* if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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University of Toronto-Mississauga
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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$
$$q_0 \sqcup q_1 \ 1 \ R$$
$$q_1 \sqcup q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
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q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

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For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



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#17 Page 6 of 12

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#17 Page 8 of 12

Q8. (5 points)

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#17 Page 10 of 12

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#18 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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#18 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

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(b) (6 points)

Prove that $3SAT \leq_p L$.



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#18 Page 10 of 12

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#19 Page 10 of 12

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(a) (1 point)

Give an example of a set A of natural numbers which is both c.e. and co-c.e.
(co-c.e. means the complement of a c.e. set).

(b) (1 point)

Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)

Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

For A, C above: Is $A \leq_T C$? Justify your answer.



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#20 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?



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#20 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -*clique* if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#20 Page 10 of 12

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#20 Page 12 of 12

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University of Toronto-Mississauga
CSC363 Final Examination
Computational Complexity and Computability Winter 2023

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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$
$$q_0 \ \sqcup \ q_1 \ 1 \ R$$
$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

(c) (3 points)

Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

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#21 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

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#21 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

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2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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$$f(x) = \text{-----}$$

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.
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(a) (1 point)

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(d) (2 points)

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(e) (2 points)

For A, C above: Is $A \leq_T C$? Justify your answer.



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#22 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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Q7. Let A, B , and C be three sets of integers.

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(a) (4 points)

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?



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#22 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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#23 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.
- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.
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Q6. Answer each of the following

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Give an example (with proof) of a problem in P.

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Q7. Let A, B , and C be three sets of integers.

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Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

- (c) A set is recognizable if and only if it is c.e. (computably enumerable).

- (d) $P \subseteq NP$.

- (e) Every Boolean formula is equivalent to a 3CNF formula.

- (f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

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**Q5. Answer each of the following**

(a) (1 point)

Give an example of a set A of natural numbers which is both c.e. and co-c.e.
(co-c.e. means the complement of a c.e. set).

(b) (1 point)

Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)

Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

For A, C above: Is $A \leq_T C$? Justify your answer.



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#24 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#24 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#24 Page 10 of 12

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#24 Page 12 of 12

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$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

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#25 Page 6 of 12

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#25 Page 8 of 12

Q8. (5 points)

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2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#25 Page 10 of 12

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$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

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(a) (3 points)

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q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

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#26 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.
- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.
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For A, B above: Is $A \leq_T B$? Justify your answer.

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#26 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#26 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#26 Page 10 of 12

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Q6. Answer each of the following

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$$f(x) = \text{-----}$$

(c) (3 points)

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$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



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#28 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

- (c) A set is recognizable if and only if it is c.e. (computably enumerable).

- (d) $P \subseteq NP$.

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**Q5. Answer each of the following**

(a) (1 point)

Give an example of a set A of natural numbers which is both c.e. and co-c.e.
(co-c.e. means the complement of a c.e. set).

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Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

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#28 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A , B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A , B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#28 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#28 Page 10 of 12

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#28 Page 12 of 12

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University of Toronto-Mississauga
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$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
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q_1		

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#29 Page 6 of 12

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#29 Page 8 of 12

Q8. (5 points)

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#30 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

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(a) (4 points)

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#30 Page 8 of 12

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5. Exam type: Closed book. No calculator.
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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

(c) (3 points)

Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \sqcup q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

- (c) A set is recognizable if and only if it is c.e. (computably enumerable).

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**Q5. Answer each of the following**

(a) (1 point)

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(co-c.e. means the complement of a c.e. set).

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Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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#32 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#32 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#32 Page 10 of 12

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University of Toronto-Mississauga
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Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



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#33 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

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- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.
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(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

For A, C above: Is $A \leq_T C$? Justify your answer.



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#33 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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#33 Page 8 of 12

Q8. (5 points)

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Let L be the following language:

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(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#33 Page 10 of 12

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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#34 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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#34 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#35 Page 6 of 12

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#35 Page 8 of 12

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University of Toronto-Mississauga
CSC363 Final Examination
Computational Complexity and Computability Winter 2023

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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$
$$q_0 \sqcup q_1 \ 1 \ R$$
$$q_1 \sqcup q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

(c) (3 points)

Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \sqcup q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



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#36 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.
- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.
- (c) A set is recognizable if and only if it is c.e. (computably enumerable).
- (d) $P \subseteq NP$.
- (e) Every Boolean formula is equivalent to a 3CNF formula.
- (f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.
- (g) The Halting Problem is an NP-complete problem.

**Q5. Answer each of the following**

(a) (1 point)

Give an example of a set A of natural numbers which is both c.e. and co-c.e.
(co-c.e. means the complement of a c.e. set).

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For A, B above: Is $A \leq_T B$? Justify your answer.

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#36 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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i. Prove that $D \leq_m E$.

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#36 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#36 Page 10 of 12

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#37 Page 6 of 12

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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#37 Page 8 of 12

Q8. (5 points)

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$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Prove that $3SAT \leq_p L$.



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Q8. (5 points)

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Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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University of Toronto-Mississauga
CSC363 Final Examination
Computational Complexity and Computability Winter 2023

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6. Materials allowed: No additional materials are allowed.



Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.
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(a) (1 point)

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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#40 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

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Prove that: $B \leq_p D$.

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#40 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

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(b) (6 points)

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#40 Page 10 of 12

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Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

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#41 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

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(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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#41 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

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(b) (6 points)

Prove that $3SAT \leq_p L$.



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#41 Page 10 of 12

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#42 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

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(a) (1 point)

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#42 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

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Q8. (5 points)

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Prove that $3SAT \leq_p L$.



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#42 Page 10 of 12

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#43 Page 6 of 12

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Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -*clique* if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#43 Page 10 of 12

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University of Toronto-Mississauga
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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \sqcup \ q_1 \ 1 \ R$$

$$q_1 \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \sqcup q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

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#44 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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#44 Page 8 of 12

Q8. (5 points)

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$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#44 Page 10 of 12

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#45 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

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#45 Page 8 of 12

Q8. (5 points)

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#46 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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Q7. Let A, B , and C be three sets of integers.

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#47 Page 6 of 12

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Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?



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#47 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -clique if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#47 Page 10 of 12

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University of Toronto-Mississauga
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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



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#48 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

- (c) A set is recognizable if and only if it is c.e. (computably enumerable).

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(a) (1 point)

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#48 Page 6 of 12

Q6. Answer each of the following

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#48 Page 8 of 12

Q8. (5 points)

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#48 Page 10 of 12

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q_1		

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

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(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

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#49 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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#49 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Prove that $3SAT \leq_p L$.



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#49 Page 10 of 12

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#49 Page 12 of 12

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University of Toronto-Mississauga
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#50 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

For A, C above: Is $A \leq_T C$? Justify your answer.



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#51 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?



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#51 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -*clique* if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#51 Page 10 of 12

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University of Toronto-Mississauga
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$$q_0 \ 1 \ q_0 \ 1 \ R$$
$$q_0 \ \sqcup \ q_1 \ 1 \ R$$
$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



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#52 Page 6 of 12

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#52 Page 8 of 12

Q8. (5 points)

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Q9. (2 points)

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#53 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

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#53 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

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#53 Page 10 of 12

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#54 Page 10 of 12

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(a) (1 point)

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(co-c.e. means the complement of a c.e. set).

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Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

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#55 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#55 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#55 Page 10 of 12

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University of Toronto-Mississauga
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$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \sqcup q_1 \ 1 \ R$$

$$q_1 \sqcup q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

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q_1		

(b) (2 points)

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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#56 Page 6 of 12

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#57 Page 6 of 12

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Q8. (5 points)

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#57 Page 10 of 12

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

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(a) (1 point)

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(co-c.e. means the complement of a c.e. set).

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Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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#59 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#59 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#59 Page 10 of 12

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$$q_0 \ 1 \ q_0 \ 1 \ R$$

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(a) (3 points)

Complete the following table to describe a possible transition function δ :

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q_1		

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#60 Page 6 of 12

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#60 Page 8 of 12

Q8. (5 points)

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Q9. (2 points)

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#61 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

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#61 Page 8 of 12

Q8. (5 points)

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Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

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- (d) $P \subseteq NP$.

- (e) Every Boolean formula is equivalent to a 3CNF formula.

- (f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

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**Q5. Answer each of the following**

(a) (1 point)

Give an example of a set A of natural numbers which is both c.e. and co-c.e.
(co-c.e. means the complement of a c.e. set).

(b) (1 point)

Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)

Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

For A, C above: Is $A \leq_T C$? Justify your answer.



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#63 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?



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#63 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -*clique* if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#63 Page 10 of 12

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University of Toronto-Mississauga
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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

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#64 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

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#64 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

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#64 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

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2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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Prove the following statement: Every finite language is decidable.

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#65 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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Q8. (5 points)

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(b) (6 points)

Prove that $3SAT \leq_p L$.



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Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Prove that $3SAT \leq_p L$.



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6. Materials allowed: No additional materials are allowed.



Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

(c) (3 points)

Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

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Q5. Answer each of the following

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Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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#67 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

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Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

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Prove that $3SAT \leq_p L$.



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#68 Page 6 of 12

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#68 Page 8 of 12

Q8. (5 points)

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Prove that $3SAT \leq_p L$.



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Q8. (5 points)

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University of Toronto-Mississauga
CSC363 Final Examination
Computational Complexity and Computability Winter 2023

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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \sqcup q_1 \ 1 \ R$$

$$q_1 \sqcup q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

(c) (3 points)

Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \sqcup q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

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#71 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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#71 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#71 Page 10 of 12

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#72 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

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#72 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

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Prove that $3SAT \leq_p L$.



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#72 Page 10 of 12

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#73 Page 8 of 12

Q8. (5 points)

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University of Toronto-Mississauga
CSC363 Final Examination
Computational Complexity and Computability Winter 2023

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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$
$$q_0 \ \sqcup \ q_1 \ 1 \ R$$
$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

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Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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(d) (2 points)

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Q6. Answer each of the following

(a) (3 points)

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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#76 Page 8 of 12

Q8. (5 points)

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Prove that $3SAT \leq_p L$.



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Q8. (5 points)

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#78 Page 6 of 12

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Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -clique if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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University of Toronto-Mississauga
CSC363 Final Examination
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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

(c) (3 points)

Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



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#79 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

- (c) A set is recognizable if and only if it is c.e. (computably enumerable).

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(a) (1 point)

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For A, B above: Is $A \leq_T B$? Justify your answer.

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#79 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

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#79 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

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(b) (6 points)

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#79 Page 10 of 12

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

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For A, B above: Is $A \leq_T B$? Justify your answer.

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#80 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#80 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#80 Page 10 of 12

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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#81 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

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#81 Page 8 of 12

Q8. (5 points)

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Prove that $3SAT \leq_p L$.



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#82 Page 6 of 12

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(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?



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Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -*clique* if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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University of Toronto-Mississauga
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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

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#83 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

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#83 Page 8 of 12

Q8. (5 points)

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2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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q_1		

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

For A, C above: Is $A \leq_T C$? Justify your answer.



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#84 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#84 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#84 Page 10 of 12

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Prove the following statement: Every finite language is decidable.

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#85 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

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#85 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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Q7. Let A, B , and C be three sets of integers.

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#85 Page 8 of 12

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#85 Page 10 of 12

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For A, C above: Is $A \leq_T C$? Justify your answer.



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#86 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -*clique* if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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University of Toronto-Mississauga
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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

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q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

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For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



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#87 Page 6 of 12

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#87 Page 8 of 12

Q8. (5 points)

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

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#88 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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#88 Page 8 of 12

Q8. (5 points)

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$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#88 Page 10 of 12

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**Q5. Answer each of the following**

(a) (1 point)

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(co-c.e. means the complement of a c.e. set).

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Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

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#90 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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#90 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -clique if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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University of Toronto-Mississauga
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$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

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q_1		

(b) (2 points)

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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#91 Page 6 of 12

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#91 Page 8 of 12

Q8. (5 points)

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(b) (6 points)

Prove that $3SAT \leq_p L$.



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#92 Page 6 of 12

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#92 Page 8 of 12

Q8. (5 points)

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(b) (6 points)

Prove that $3SAT \leq_p L$.



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#92 Page 10 of 12

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For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

- (c) A set is recognizable if and only if it is c.e. (computably enumerable).

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(a) (1 point)

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(co-c.e. means the complement of a c.e. set).

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Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

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#94 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#94 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -*clique* if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#94 Page 10 of 12

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(a) (3 points)

Complete the following table to describe a possible transition function δ :

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q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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#95 Page 6 of 12

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#95 Page 8 of 12

Q8. (5 points)

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2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#96 Page 6 of 12

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Q7. Let A, B , and C be three sets of integers.

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#96 Page 8 of 12

Q8. (5 points)

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(b) (6 points)

Prove that $3SAT \leq_p L$.



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#96 Page 10 of 12

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



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#98 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

- (c) A set is recognizable if and only if it is c.e. (computably enumerable).

- (d) $P \subseteq NP$.

- (e) Every Boolean formula is equivalent to a 3CNF formula.

- (f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

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**Q5. Answer each of the following**

(a) (1 point)

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(co-c.e. means the complement of a c.e. set).

(b) (1 point)

Give an example of a set B of natural numbers which is not c.e. but co-c.e.

(c) (1 point)

Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

For A, C above: Is $A \leq_T C$? Justify your answer.



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#98 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?



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#98 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -clique if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

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(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

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#99 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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(b) (3 points)

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#99 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

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2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#99 Page 10 of 12

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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#100 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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#100 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#100 Page 10 of 12

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Q4. Write True or False with a brief justification (2 points each)

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#101 Page 6 of 12

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(b) (6 points)

Prove that $3SAT \leq_p L$.



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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \sqcup \ q_1 \ 1 \ R$$

$$q_1 \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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$$gn(q_0 \sqcup q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

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Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

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Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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#102 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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University of Toronto-Mississauga
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Q3. (3 points)

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#103 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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#103 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#103 Page 10 of 12

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University of Toronto-Mississauga
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#104 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

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#104 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

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Prove that $3SAT \leq_p L$.



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#104 Page 10 of 12

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#105 Page 6 of 12

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#105 Page 8 of 12

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CSC363 Final Examination
Computational Complexity and Computability Winter 2023

First name (please write as legibly as possible within the boxes)

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4. *Do NOT detach any paper from this exam.*
5. Exam type: Closed book. No calculator.
6. Materials allowed: No additional materials are allowed.



Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

(c) (3 points)

Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

- (c) A set is recognizable if and only if it is c.e. (computably enumerable).

- (d) $P \subseteq NP$.

- (e) Every Boolean formula is equivalent to a 3CNF formula.

- (f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

- (g) The Halting Problem is an NP-complete problem.

**Q5. Answer each of the following**

(a) (1 point)

Give an example of a set A of natural numbers which is both c.e. and co-c.e.
(co-c.e. means the complement of a c.e. set).

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For A, B above: Is $A \leq_T B$? Justify your answer.

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Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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i. Prove that $D \leq_m E$.

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#106 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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1. C has k -many elements (vertices).
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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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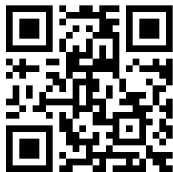
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Prove that $3SAT \leq_p L$.



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Q8. (5 points)

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Prove that $3SAT \leq_p L$.



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Prove that $3SAT \leq_p L$.



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#109 Page 10 of 12

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University of Toronto-Mississauga
CSC363 Final Examination
Computational Complexity and Computability Winter 2023

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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

(c) (3 points)

Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \sqcup q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



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#110 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

- (c) A set is recognizable if and only if it is c.e. (computably enumerable).

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For A, B above: Is $A \leq_T B$? Justify your answer.

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#110 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

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#110 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

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(b) (6 points)

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Prove the following statement: Every finite language is decidable.

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(d) (2 points)

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#111 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

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Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#111 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

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(b) (6 points)

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Q8. (5 points)

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Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -*clique* if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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University of Toronto-Mississauga
CSC363 Final Examination
Computational Complexity and Computability Winter 2023

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6. Materials allowed: No additional materials are allowed.



Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

(c) (3 points)

Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

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**Q5. Answer each of the following**

(a) (1 point)

Give an example of a set A of natural numbers which is both c.e. and co-c.e.
(co-c.e. means the complement of a c.e. set).

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Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

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#114 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?



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#114 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

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2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#114 Page 10 of 12

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	1	$\sqcup$
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q_1		

(b) (2 points)

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$$f(x) = \text{-----}$$

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

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For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



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(a) (1 point)

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

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#115 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?



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Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

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2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#115 Page 10 of 12

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#115 Page 12 of 12

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University of Toronto-Mississauga
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$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

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q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

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Q5. Answer each of the following

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(e) (2 points)

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#116 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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#116 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#116 Page 10 of 12

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#117 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

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#117 Page 6 of 12

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?



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Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -clique if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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University of Toronto-Mississauga
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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

(c) (3 points)

Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



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(co-c.e. means the complement of a c.e. set).

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(d) (2 points)

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#118 Page 6 of 12

Q6. Answer each of the following

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#118 Page 8 of 12

Q8. (5 points)

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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(a) (3 points)

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q_1		

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

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Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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#119 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

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$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#120 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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Q7. Let A, B , and C be three sets of integers.

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#120 Page 8 of 12

Q8. (5 points)

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#120 Page 10 of 12

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

For A, C above: Is $A \leq_T C$? Justify your answer.



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Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?



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#121 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -*clique* if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#121 Page 10 of 12

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University of Toronto-Mississauga
CSC363 Final Examination
Computational Complexity and Computability Winter 2023

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6. Materials allowed: No additional materials are allowed.



Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

(c) (3 points)

Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



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#122 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

- (c) A set is recognizable if and only if it is c.e. (computably enumerable).

- (d) $P \subseteq NP$.

- (e) Every Boolean formula is equivalent to a 3CNF formula.

- (f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

- (g) The Halting Problem is an NP-complete problem.

**Q5. Answer each of the following**

(a) (1 point)

Give an example of a set A of natural numbers which is both c.e. and co-c.e.
(co-c.e. means the complement of a c.e. set).

(b) (1 point)

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#122 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

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#122 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -*clique* if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#122 Page 10 of 12

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q_1		

(b) (2 points)

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

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(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

For A, C above: Is $A \leq_T C$? Justify your answer.



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#123 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#123 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#123 Page 10 of 12

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#123 Page 12 of 12

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University of Toronto-Mississauga
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$$q_1 \ \sqcup \ q_h \ 1 \ R$$

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(b) (2 points)

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

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#124 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

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#124 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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Q5. Answer each of the following

(a) (1 point)

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(co-c.e. means the complement of a c.e. set).

(b) (1 point)

Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

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For A, B above: Is $A \leq_T B$? Justify your answer.

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#125 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

(c) (4 points)

Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



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1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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University of Toronto-Mississauga
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$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
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q_1		

(b) (2 points)

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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

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#126 Page 6 of 12

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#126 Page 8 of 12

Q8. (5 points)

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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Prove the following statement: Every finite language is decidable.

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#127 Page 6 of 12

Q6. Answer each of the following

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#127 Page 8 of 12

Q8. (5 points)

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(b) (6 points)

Prove that $3SAT \leq_p L$.



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#127 Page 10 of 12

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#128 Page 10 of 12

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Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



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#129 Page 4 of 12

Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

- (c) A set is recognizable if and only if it is c.e. (computably enumerable).

- (d) $P \subseteq NP$.

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- (f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

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Q5. Answer each of the following

(a) (1 point)

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(co-c.e. means the complement of a c.e. set).

(b) (1 point)

Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

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#129 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

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#129 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#129 Page 10 of 12

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University of Toronto-Mississauga
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$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \sqcup \ q_1 \ 1 \ R$$

$$q_1 \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

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#130 Page 6 of 12

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#130 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

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Q10. (3 points)

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2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#131 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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#131 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

(c) (3 points)

Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

- (a) Every recognizable Language is decidable.

- (b) Let A, B be two recognizable languages. Then, the union $A \cup B$ is also a recognizable language.

- (c) A set is recognizable if and only if it is c.e. (computably enumerable).

- (d) $P \subseteq NP$.

- (e) Every Boolean formula is equivalent to a 3CNF formula.

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**Q5. Answer each of the following**

(a) (1 point)

Give an example of a set A of natural numbers which is both c.e. and co-c.e.
(co-c.e. means the complement of a c.e. set).

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Give an example of a set B of natural numbers which is not c.e. but co-c.e.

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Give an example of a set C of natural numbers which is neither c.e. nor co-c.e.

(d) (2 points)

For A, B above: Is $A \leq_T B$? Justify your answer.

(e) (2 points)

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#133 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

(b) (2 points)

Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

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#133 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

Hint: Prove the contrapositive of the statement.



Q11. Recall that an undirected graph $G = (V, E)$ is said to have a k -clique if there exists $C \subseteq V$ such that:

1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#133 Page 10 of 12

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University of Toronto-Mississauga
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Q2. (3 points)

Prove the following statement: Every finite language is decidable.

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Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

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For A, B above: Is $A \leq_T B$? Justify your answer.

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#134 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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i. Prove that $D \leq_m E$.

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#134 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

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1. C has k -many elements (vertices).
2. Every two vertices in C are connected by an edge in the graph.

Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#135 Page 8 of 12

Q8. (5 points)

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(a) (3 points)

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(b) (6 points)

Prove that $3SAT \leq_p L$.



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#135 Page 10 of 12

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#136 Page 6 of 12

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#136 Page 8 of 12

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Prove that $3SAT \leq_p L$.



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#136 Page 10 of 12

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Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

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$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

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#137 Page 6 of 12

Q6. Answer each of the following

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Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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#137 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

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(b) (6 points)

Prove that $3SAT \leq_p L$.



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#138 Page 6 of 12

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Q8. (5 points)

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Prove that $3SAT \leq_p L$.



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University of Toronto-Mississauga
CSC363 Final Examination
Computational Complexity and Computability Winter 2023

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6. Materials allowed: No additional materials are allowed.



Q1. Consider the following Turing program:

$$q_0 \ 1 \ q_0 \ 1 \ R$$

$$q_0 \ \sqcup \ q_1 \ 1 \ R$$

$$q_1 \ \sqcup \ q_h \ 1 \ R$$

(a) (3 points)

Complete the following table to describe a possible transition function δ :

	1	$\sqcup$
q_0		$q_1, 1, R$
q_1		

(b) (2 points)

Suppose that the convention is to input x (natural numbers) as $x+1$ consecutive 1's on the tape, and the output is the number of 1's left on the tape after the machine halts (reaches the halting state q_h). According to that input/output convention, which function does our Turing machine compute? (fill the gap below)

$$f(x) = \text{-----}$$

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Let's give our machine a Gödel number (encode it as a natural number). Let gn denote the Gödel number function (from the set of strings made of machine characters to $\mathbb{N}$). Suppose $gn(1) = 1$, $gn(\sqcup) = 2$, $gn(R) = 3$, $gn(L) = 4$, $gn(q_h) = 5$, and for every natural number i , $gn(q_i) = 6 + i$. This implies that:

$$gn(q_0 \ \sqcup \ q_1 \ 1 \ R) = 2^6 3^2 5^7 7^1 11^3.$$

Compute the Gödel number of the Turing program above (you can leave your final answer factored into primes since you do not have a calculator).



Q2. (3 points)

Prove the following statement: Every finite language is decidable.

Q3. (3 points)

Recall that, for a natural number e , φ_e denotes the partial recursive function equivalent to the Turing machine whose Gödel number is e .

Prove that:

For every partial recursive function g , the set $\{x \in \mathbb{N} : g = \varphi_x\}$ is infinite. In other words, every partial recursive function has an infinite number of machines that compute it.



Q4. Write True or False with a brief justification (2 points each)

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- (d) $P \subseteq NP$.

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- (f) There exists two languages A, B such that $A \leq_p B$ but not $A \leq_T B$.

- (g) The Halting Problem is an NP-complete problem.

**Q5. Answer each of the following**

(a) (1 point)

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#141 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

Give an example (with proof) of a problem in P.

(b) (3 points)

Give three examples of problems which are known to be NP-complete.



Q7. Let A, B , and C be three sets of integers.

Let $D = \{3x : x \in A\} \cup \{3x + 1 : x \in B\} \cup \{3x + 2 : x \in C\}$.

(a) (4 points)

Prove that: $B \leq_p D$.

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Do we have that $A \leq_p D$ and $C \leq_p D$?

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Suppose that there exists a set E to which all of A, B , and C are m-reducible (i.e., $A \leq_p E$, $B \leq_p E$, and $C \leq_p E$).

i. Prove that $D \leq_m E$.

ii. Does your proof automatically show that $D \leq_p E$?



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#141 Page 8 of 12

Q8. (5 points)

Let $g(n)$ be a polynomial of degree k . Prove that $g(n)$ is in $O(n^k)$.

Q9. (2 points)

Describe briefly how you can get an NP-complete set of only even natural numbers.

Q10. (3 points)

Show that if there is a language L such that $L \in \text{co-NP}$ and $L \notin \text{NP}$, then $\text{P} \neq \text{NP}$.

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1. C has k -many elements (vertices).
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Let L be the following language:

$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

Prove that $L \in NP$ by using a verifier in $O(n^2)$.

(b) (6 points)

Prove that $3SAT \leq_p L$.



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#142 Page 6 of 12

Q6. Answer each of the following

(a) (3 points)

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Q7. Let A, B , and C be three sets of integers.

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#142 Page 8 of 12

Q8. (5 points)

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$$L = \{(G, k) : G \text{ is an undirected graph which has a } k\text{-clique}\}.$$

(a) (3 points)

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(b) (6 points)

Prove that $3SAT \leq_p L$.



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Q8. (5 points)

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Prove that $3SAT \leq_p L$.



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Prove that $3SAT \leq_p L$.



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